| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 12. | (b) | | | $f = c/\lambda = 3 \times 10^8/278 \times 10^{-9} = 1.079 \times 10^{15}$ (Hz) (1)<br><br>$E = hf \quad \therefore E = 6.63 \times 10^{-34} \times 1.079 \times 10^{15}$ J<br>$= 7.154 \times 10^{-19}$ J (per molecule)<br><br>$\therefore$ per mole $7.154 \times 10^{-19} \times 6.02 \times 10^{23}$ J mol$^{-1}$<br>$= 4.307 \times 10^5$ J mol$^{-1}$ (1)<br>$= 431$ kJ mol$^{-1}$ | | | 2 | 2 | 2 | |
| | (c) | | | As the group is descended the bond energies decrease and the wavelengths increase / astatine is below iodine in the Periodic Table / $\lambda_{max} > 400$nm (1)<br><br>coloured gas linked with the visible region (1) | | | 2 | 2 | | |
| | | | | **Question 12 total** | **1** | **7** | **5** | **13** | **3** | **4** |